

Chapter 10: Prenatal Care — Revision Notes

Williams Obstetrics 26e, pp. 461–514 of the PDF. High-yield numbers are in **bold**. The seven tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 4 figures are included from the PDF.

Big picture: Prenatal care = coordinated medical care + continuous risk assessment + psychosocial support, ideally starting before pregnancy and extending through the postpartum and interpregnancy period. No prenatal care carries a **5-fold** higher risk of maternal death.

1. Prenatal Care in the United States

- Only **1.6%** of US women giving birth (2016) had **no prenatal care**.
 - Inadequate or no care: **10% African-American, 7.7% Hispanic**; higher in adolescents, especially **<15 years**.

Prenatal care effectiveness

- Maternal deaths fell from **690/100,000 (1920) to 50/100,000 (1955)**, helped by prenatal care.
- **No prenatal care → 5-fold ↑ maternal death** (1998–2005 surveillance).
- In ~29 million births, preterm birth, stillbirth, neonatal and infant death rose **linearly with ↓ prenatal care use**.
- Parkland: preterm births fell with any prenatal care use by indigent women. In diabetics, adherence → fewer NICU admissions.
- **Group prenatal care** is acceptable and effective, with **better outcomes** than individual care (Ickovics, 2016). **Childbirth education classes** also improve outcomes.

2. Diagnosis of Pregnancy

Symptoms and signs

- **Amenorrhea** is not a reliable sign until **≥10 days after expected menses**.
- Bleeding in the 1st month may be **implantation bleeding**, but any 1st-trimester bleeding should prompt evaluation for an abnormal pregnancy.
- **Quickening (fetal movement felt):**
 - Multipara: **16–18 weeks**
 - Primigravida: about **2 weeks later**
 - An examiner can feel movements at about **20 weeks**.

Pregnancy tests (hCG)

- Made by **syncytiotrophoblast**; rises exponentially in the 1st trimester.
- hCG and LH share the **same receptor**. Main function: **maintains the corpus luteum**, the principal progesterone source for the first **6 weeks**.
- Detectable in serum/urine **8–9 days after ovulation**.
- **Doubling time 1.4–2.0 days**. Lower levels rise faster than higher ones.
- **Peak at 60–70 days**, then falls slowly to a **plateau at ~16 weeks**.
- **Structure:** glycoprotein heterodimer. The **α-subunit is identical to LH, FSH and TSH**; the **β-subunit is distinct** → assays use β-specific antibodies. Serum detection limit **≤1.0 mIU/mL**.

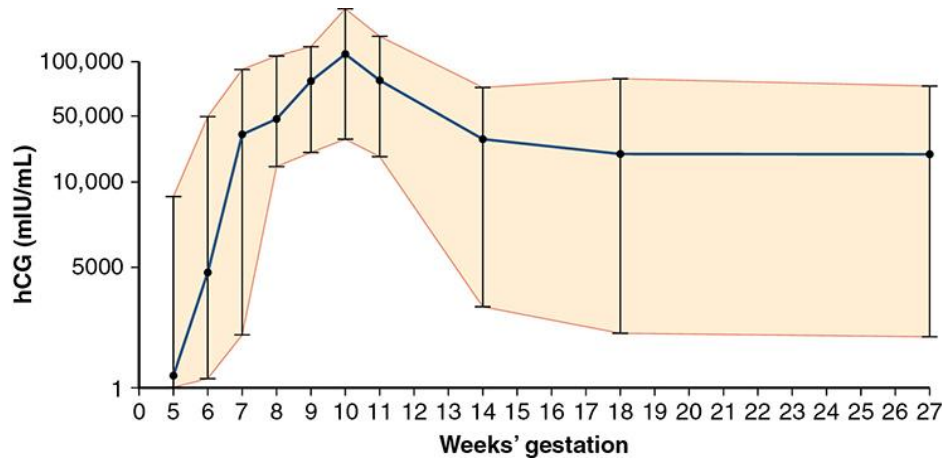


Figure 10-1. Serum hCG across normal pregnancy (mean, 95% CI): steep rise, peak ~100,000 mIU/mL at ~10 wk, then plateau from ~16 wk; the normal range is very wide.

False-positive hCG (rare)

- **Heterophilic antibodies** (most common): bind the animal-derived test antibodies; more likely in women who **work with animals** → use alternative lab methods.
- Molar pregnancy and gestational trophoblastic neoplasia (raised hCG).
- Other causes without pregnancy:
 1. **Exogenous hCG** (weight-loss injections)
 2. **Renal failure** (↓ clearance)
 3. **Physiological pituitary hCG**
 4. **hCG-producing tumors**: GI tract, ovary, bladder, lung

Home pregnancy tests

- >60 kits in the US; many are **less accurate than advertised**.
- Diagnosing **95%** of pregnancies at the missed period needs a detection limit of **12.5 mIU/mL**; only 1 brand achieved this.
- At 100 mIU/mL only **44%** of brands were clearly positive → only **~15%** of pregnancies detected at the missed period.

Sonographic recognition

- **Transvaginal sonography** establishes gestational age and location.
- **Gestational sac** is the first sign: seen at **4–5 weeks**.
- **Pseudosac** (fluid in the cavity with ectopic pregnancy) sits in the **midline**; a true sac implants **eccentrically**.
- Early IUP signs:
 - **Intradecidual sign**: anechoic center with a **single echogenic rim**
 - **Double decidual sign**: **two concentric echogenic rings** (decidua parietalis + decidua capsularis)
- **Yolk sac** within the sac **confirms an intrauterine pregnancy**; seen by the **middle of week 5**.
- **Embryo** with **cardiac motion** next to the yolk sac after **6 weeks**.
- Equivocal findings = **pregnancy of unknown location (PUL)** → serial hCG + TVS.

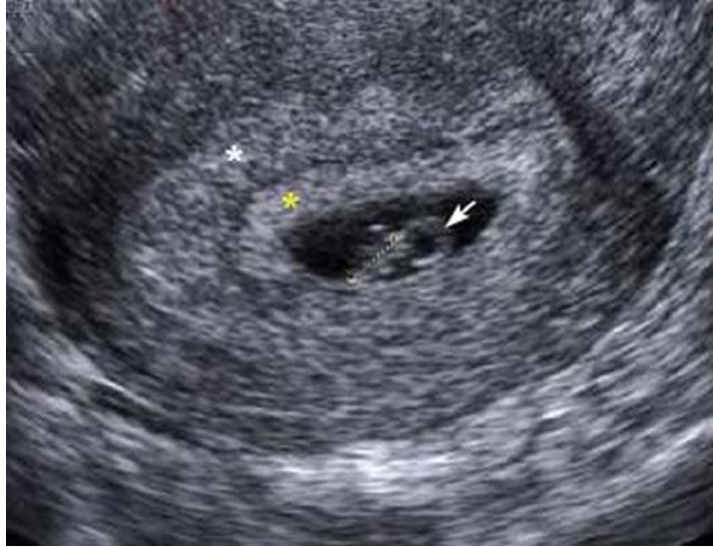


Figure 10-2. Early IUP on TVS: double decidual sign (white asterisk = decidua parietalis, yellow = decidua capsularis), yolk sac (arrow), embryo CRL between calipers.

3. Initial Prenatal Evaluation

- Goals:
 1. Define the **health status** of mother and fetus
 2. Estimate **gestational age**
 3. Plan continued obstetrical care

Table 10-1. Typical Components of Routine Prenatal Care

	Text Referral	First Visit	15–20 wk	24–28 wk	29–41 wk
History					
Complete	Chap. 10, p. 179	•			
Updated			•	•	•
Physical Examination					
Complete	Chap. 10, p. 180	•			
Blood pressure	Chap. 40, p. 688	•	•	•	•
Maternal weight	Chap. 10, p. 181	•	•	•	•
Pelvic/cervical examination	Chap. 10, p. 180	•			
Fundal height	Chap. 10, p. 180	•	•	•	•
Fetal heart rate/fetal position	Chap. 10, p. 182	•	•	•	•
Laboratory Tests					
Hematocrit or hemoglobin	Chap. 59, p. 1048	•		•	
Blood type and Rh factor	Chap. 18, p. 353	•			
Antibody screen	Chap. 18, p. 353	•		A	
Pap smear screening	Chap. 66, p. 1164	•			
Glucose tolerance test	Chap. 60, p. 1079			•	
Fetal aneuploidy screening	Chap. 17, p. 335	B ^a and/or	B		
Neural-tube defect screening	Chap. 17, p. 338		B		
Cystic fibrosis screening	Chap. 17, p. 342	B or	B		
Urine protein assessment	Chap. 4, p. 68	•			
Urine culture	Chap. 56, p. 996	•			
Rubella serology	Chap. 67, p. 1190	•			
Syphilis serology	Chap. 68, p. 1208	•			C
Gonococcal screening	Chap. 68, p. 1211	D			D
Chlamydial screening	Chap. 68, p. 1212	•			C
Hepatitis B serology	Chap. 58, p. 1037	•			D
HIV serology	Chap. 68, p. 1219	B			D
Group B streptococcus culture	Chap. 67, p. 1195				E
Tuberculosis screening	Chap. 54, p. 966	F			

^aFirst-trimester aneuploidy screening may be offered between 10 and 14 weeks. A Performed at 28 weeks, if indicated. B Test should be offered. C High-risk women should be retested at the beginning of the third trimester. D High-risk women should be screened at the first prenatal visit and again in the third trimester. E Rectovaginal culture should be obtained between 35 and 37 weeks. F High-risk women should be screened at the first prenatal visit. HIV = human immunodeficiency virus.

Definitions (prenatal record)

Term	Definition
Nulligravida	Not pregnant now and never pregnant
Gravida	Pregnant now or in the past, whatever the outcome (primigravida → multigravida)
Nullipara	Never completed a pregnancy beyond 20 weeks (may have had abortions or an ectopic)
Primipara	Delivered once of fetus(es), alive or dead, at ≥20 weeks
Multipara	Two or more pregnancies reaching ≥20 weeks

- Parity counts **pregnancies** reaching 20 wk: **twins do not raise it**, and **stillbirth does not lower it**.
- The old **500-g** threshold for parity is now controversial (neonates <500 g now survive).
- **Four-digit notation (T-P-A-L)**: term – preterm – abortuses (<20 wk) – living. E.g. **para 2–1–0–3** = 2 term, 1 preterm, 0 abortuses, 3 living children.

Memory hook: "Texas Power And Light" = Term, Preterm, Abortions, Living.

Pregnancy duration and dating

- **~280 days (40 weeks)** from the first day of the LMP.
- **Naegle rule:** LMP **+7 days –3 months**. E.g. LMP Oct 5 → EDD **July 12**.
- **First-trimester ultrasound is the most accurate** way to establish or confirm gestational age (ACOG, AIUM, SMFM). ART pregnancies are dated by **embryo age / transfer date**.
- **Trimesters:** 1st through **14 wk**, 2nd through **28 wk**, 3rd through **42 wk** (three periods of 14 weeks). **"Fourth trimester" = the 12 weeks after delivery**.
- Clinically, use **completed weeks + days** (e.g. 33⁴/₇ wk or 33 + 4), not trimesters ("third-trimester bleeding" is imprecise).

Previous and current health status

- Many obstetrical complications **recur** → detailed prior-pregnancy history.
- Menstrual history (irregular menses → dating less accurate) and contraception (some methods favor **ectopic** implantation after failure).

Psychosocial screening

- Screen **all** women, **at least once per trimester**: barriers to care, communication, nutrition, unstable housing, desire for pregnancy, safety/**IPV**, depression, stress, tobacco/alcohol/drugs.
- Universal screening → fewer **preterm and low-birthweight** newborns.

Cigarette smoking

- Prevalence in pregnancy: **12–13%** (2000–2010) → **7.2% (2016)**.
- **E-cigarettes/vaping:** 0.6–15%; 7% used them before conception and postpartum, 1.4% in the last 3 months of pregnancy.
- Smoking → subfertility, miscarriage, stillbirth, low birthweight, preterm delivery. **Previa, abruption and PROM ↑ 2-fold**.
- USPSTF: offer counseling and intervention at the **first and every** prenatal visit. Quitting at **any** stage helps, best early or preconceptionally.
- **Person-to-person psychosocial interventions** beat simple advice → the **5 A's** (≤15 minutes).

Table 10-2. Five A's of Smoking Cessation

Step	Action
ASK	about smoking at the first and subsequent prenatal visits.
ADVISE	with clear, strong statements that explain the risks of continued smoking to the woman, fetus, and newborn.
ASSESS	the patient's willingness to attempt cessation.
ASSIST	with pregnancy-specific, self-help smoking cessation materials. Offer a direct referral to the smokers' quit line (1-800-QUIT NOW) to provide ongoing counseling and support.
ARRANGE	to track smoking abstinence progress at subsequent visits.

Adapted from American College of Obstetricians and Gynecologists, 2020b; Fiore, 2008.

- **Nicotine replacement:** not adequately evaluated for safety or efficacy in pregnancy; trials conflicting (**SNAP:** possible better infant development; **SNIPP:** no difference in quit rates or birthweight). Bupropion: similar preliminary results. **Financial incentives** help.
- ACOG: if NRT is used, do so **under close supervision** after weighing smoking vs NRT risks.

Alcohol

- Potent teratogen → **fetal alcohol spectrum disorders**. **Fetal alcohol syndrome** = **growth restriction + facial abnormalities + CNS dysfunction**. Prevalence of FASD **11–50 per 1000**.
- **Abstain completely** when pregnant or planning pregnancy. **12%** of pregnant women drink (2015–2017).

Illicit drugs

- **~10%** of fetuses are exposed. Sequelae of chronic heavy use: **FGR, low birthweight, neonatal withdrawal**. Marijuana effects less convincing.
- **Heroin** → **methadone maintenance** in a registered program: start **10–30 mg orally daily**, titrate. Careful taper may suit some. **Buprenorphine ± naloxone** by credentialed physicians.

Intimate-partner violence

- Assault and coercion: physical, psychological, sexual, isolation, stalking, deprivation, intimidation, **reproductive coercion**.
- **Prevalence in pregnancy 4–8%:** more common than any condition found by routine screening **except possibly preeclampsia**.
- Linked to **preterm delivery, FGR, perinatal death**.
- **Screen at the first visit, at least once per trimester, and postpartum**, privately (self-administered or computerized screens work as well).
- Know state reporting laws; involve social services; National Domestic Violence Hotline **1-800-799-SAFE (7233)**.

Clinical evaluation

- Full physical and pelvic exam at the first visit.
- **Cervix:** bluish-red passive hyperemia (characteristic, not diagnostic); **nabothian cysts** may be prominent; normally closed except at the external os.
- **Pap test** per guidelines; chlamydia/gonorrhea specimens when indicated.
- **Bimanual exam:** cervical consistency, length, dilation; uterine and adnexal size; **bony pelvis**; vaginal/perineal anomalies. Rectal exam only for symptoms.

Gestational age by uterine size

Uterine size	Gestation
Small orange	~6 weeks
Large orange	~8 weeks
Grapefruit	~12 weeks

Laboratory tests

- **Initial bloods:** CBC, **blood type + Rh**, **antibody screen**.
- **Universal HIV testing** (opt-out: the woman may decline; record it).
- **All women:** hepatitis B, syphilis, **rubella immunity**.
- **Urine culture** recommended (treating bacteriuria ↓ symptomatic UTI). Routine urinalysis after the first visit is **unnecessary without hypertension**.

Cervical infections

- **Chlamydia:** isolated in **2–13%** of pregnant women.
 - **Screen all** at the first visit; retest in the **3rd trimester** if at risk.
 - Risk factors: unmarried, new or multiple partners, **age <25**, inner-city residence, other STDs, little or no prenatal care.
 - **Test of cure 3–4 weeks** after treatment.
- **Gonorrhea:** cervicitis/urethritis; rarely septic arthritis.
 - Screen women **with risk factors or in high-prevalence areas** at the first visit and again in the 3rd trimester.
 - Treat for gonorrhea **plus presumed chlamydia**; test of cure afterward.

Pregnancy risk assessment

- "High-risk pregnancy" is too vague; use a **specific diagnosis** where possible.

Table 10-3. Conditions for Which Maternal-Fetal Medicine Consultation May Be Beneficial

Medical History and Conditions

- Cardiac disease—moderate to severe disorders
- Diabetes mellitus with evidence of end-organ damage or uncontrolled hyperglycemia
- Family or personal history of genetic abnormalities
- Hemoglobinopathy
- Chronic hypertension if uncontrolled or associated with renal or cardiac disease
- Renal insufficiency if associated with significant proteinuria (**≥500 mg/24 hour**), serum creatinine **≥1.5 mg/dL**, or hypertension
- Pulmonary disease if severe restrictive or obstructive, including severe asthma
- Human immunodeficiency virus infection
- Prior pulmonary embolus or deep-vein thrombosis
- Severe systemic disease, including autoimmune conditions
- Bariatric surgery
- Epilepsy if poorly controlled or requires more than one anticonvulsant
- Cancer, especially if treatment is indicated in pregnancy

Obstetrical History and Conditions

- CDE (Rh) or other blood group alloimmunization (excluding ABO, Lewis)
- Prior or current fetal structural or chromosomal abnormality
- Desire or need for prenatal diagnosis or fetal therapy
- Periconceptional exposure to known teratogens
- Infection with or exposure to organisms that cause congenital infection
- Higher-order multifetal gestation
- Severe disorders of amniotic fluid volume

4. Subsequent Prenatal Visits

- **Traditional schedule:** every **4 weeks until 28 wk** → every **2 weeks until 36 wk** → **weekly** thereafter.
- Complicated pregnancies (twins, diabetes) → every **1–2 weeks**.
- 2005 expert panel: patient-specific risk assessment, **flexible visit spacing**, health promotion, standardized records, and family health up to **1 year after birth**.
- **WHO trial (~25,000 women):** low-risk women (**80%**) seen once in the 1st trimester and then at **26, 32 and 38 weeks** → median **5 visits vs 8**, with **no disadvantage**.

Prenatal surveillance

- Each visit: fetal heart rate, growth, activity; maternal **BP and weight**; ask about pain, nausea/vomiting, bleeding, **fluid leakage**, headache, visual change, dysuria.
- Late pregnancy vaginal exam: presenting part and station, pelvic capacity, ballottement (fluid volume), cervical consistency, effacement and dilation.

Fundal height

- **20–34 weeks: fundal height (cm) ≈ gestational age (wk).**
- Measure from the **top of the symphysis to the top of the fundus**, with an **empty bladder**.
- Less accurate with obesity or leiomyomas. **FGR is missed in up to ⅓** if fundal height is used alone.

Fetal heart sounds

- **Doppler: almost always audible by 10 weeks** (without maternal obesity).
- **FHR 110–160 bpm**, heard as a double sound.
- **Nonamplified stethoscope:** audible by **20 wk in 80%**, and by **22 wk in all**.

Sound	Character	Synchronous with	Source
Funic souffle	Sharp, whistling	Fetal pulse	Blood rushing through umbilical arteries
Uterine souffle	Soft, blowing	Maternal pulse	Dilated uterine vessels , heard best low on the uterus

Sonography

- Offer **at least one** prenatal ultrasound to all women. Low-risk pregnancies average **4–5** scans.
- Only for valid indications, at the lowest exposure = **ALARA** (as low as reasonably achievable).

Subsequent laboratory tests

- **Hb/Hct** (± syphilis serology where prevalent) repeated at **28–32 weeks**.
- **HIV:** retest high-risk women in the 3rd trimester, **preferably before 36 weeks**. **HBV:** high-risk women retested at delivery.
- **D (Rh)-negative, unsensitized:** repeat antibody screen at **28–29 weeks** → give **anti-D immunoglobulin** if still unsensitized.
- **GBS: vaginal + rectal culture at 35–37 weeks** in all women; intrapartum prophylaxis if positive.
 - **Empirical intrapartum prophylaxis** (no culture needed): **GBS bacteriuria, preterm labor, previous infant with invasive GBS disease**.
- **Gestational diabetes:** screen **all** women (history, clinical factors or labs). Lab testing at **24–28 weeks** is most sensitive.
- **Aneuploidy screening** offered to all: 1st trimester **10–14 wk**, 2nd trimester **15–20 wk**, or **cell-free DNA any time after 10 wk**.
- **Cystic fibrosis and spinal muscular atrophy carrier screening** offered to **all** women considering or in pregnancy. Consider **panethnic/expanded** screening; ethnicity-based screening (e.g. Tay-Sachs in Ashkenazi Jews) remains an option. All genetic screening is **optional**, ideally before pregnancy.
- **NTD screening:** traditionally **MSAFP** in the 2nd trimester → now largely replaced by the **2nd-trimester anatomy scan**.

5. Nutritional Counseling

Weight gain recommendations

- **IOM 2009:** gain is stratified by **prepregnancy BMI**, and applies to all ages, races and ethnic groups.

Table 10-4. Recommendations for Total and Rate of Weight Gain During Pregnancy

Category (BMI)	Total Weight Gain Range (lb) ^a	Weight Gain in 2nd and 3rd Trimesters, Mean in lb/wk (range)
Underweight (<18.5)	28–40	1 (1–1.3)
Normal weight (18.5–24.9)	25–35	1 (0.8–1)
Overweight (25.0–29.9)	15–25	0.6 (0.5–0.7)
Obese (≥30.0)	11–20	0.5 (0.4–0.6)

^aEmpirical recommendations for weight gain in twin pregnancies include: normal BMI, 37–54 lb; overweight women, 31–50 lb; and obese women, 25–42 lb. BMI = body mass index. Modified from the Centers for Disease Control and Prevention, 2019b; Institute of Medicine and National Research Council, 2009.

- The emphasis has shifted from low birthweight to **obesity**; the narrow range for obese women reflects interest in **lower** gains.
- Obesity → gestational hypertension, preeclampsia, GDM, **macrosomia**, cesarean. Risk ∝ weight gain.
- MFMU cohort (>29,000): **51% gained above** and **21% below** guidelines; excess gain → more hypertension and cesarean (worse with chronic hypertension).
- Normal-BMI women gaining **<25 lb**: less preeclampsia, failed induction, CPD, cesarean and LGA, but **more SGA**.

Severe undernutrition

- Experimental deficiency is unethical; famine data suggest **near starvation** is needed to clearly change outcome in otherwise healthy women (Ch 47).

Weight retention after pregnancy

- Average gain **12.9 kg (28.6 lb)** (795 women).
- Lost at **delivery ~5.4 kg (12 lb)**; next **2 weeks ~4 kg (9 lb)**; **2 wk–6 mo another 2.5 kg (5.5 lb)**.
- **Average retained: ~1 kg (2.1 lb)**.
- Excess gain is laid down as **fat** and may be retained long term. More gain → more loss. **Longer breastfeeding → less retention**.

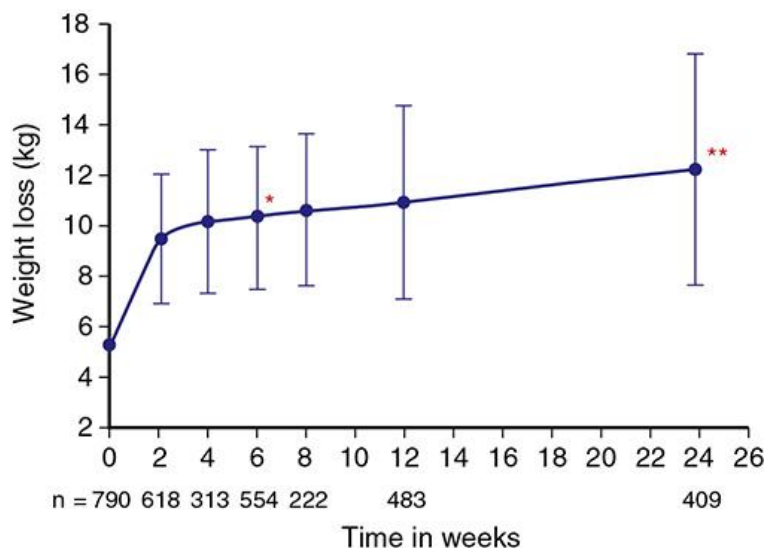


Figure 10-3. Cumulative postpartum weight loss: ~5 kg at delivery, ~9.5 kg by 2 wk, ~12 kg by 6 months; on average 1 kg is retained.

Table 10-5. Recommended Daily Dietary Allowances for Pregnant and Lactating Women

Nutrient	Pregnant	Lactating
Fat-Soluble Vitamins		
Vitamin A	770 µg	1300 µg
Vitamin D ^a	15 µg	15 µg
Vitamin E	15 mg	19 mg
Vitamin K ^a	90 µg	90 µg
Water-Soluble Vitamins		
Vitamin C	85 mg	120 mg
Thiamin	1.4 mg	1.4 mg
Riboflavin	1.4 mg	1.6 mg
Niacin	18 mg	17 mg
Vitamin B ₆	1.9 mg	2 mg
Folate	600 µg	500 µg
Vitamin B ₁₂	2.6 µg	2.8 µg
Minerals		
Calcium ^a	1000 mg	1000 mg
Sodium ^a	1.5 g	1.5 g
Potassium ^a	4.7 g	5.1 g
Iron	27 mg	9 mg
Zinc	11 mg	12 mg
Iodine	220 µg	290 µg
Selenium	60 µg	70 µg
Other		
Protein	71 g	71 g
Carbohydrate	175 g	210 g
Fiber ^a	28 g	29 g

^aRecommendations measured as adequate intake. Modified from the American Academy of Pediatrics, 2017; Institute of Medicine, 2006, 2011. Note: the text gives zinc intake as "approximately 12 mg" in pregnancy; the table lists 11 mg (12 mg is the lactation value).

- Excess self-prescribed supplements risk toxicity: **iron, zinc, selenium, vitamins A, B6, C and D.**

Calories

- Pregnancy needs **~80,000 kcal** extra, mostly in the **last 20 weeks**.
- AAP/ACOG: **+100–300 kcal/d**. IOM: **+0 / 340 / 452 kcal/d** in the 1st / 2nd / 3rd trimesters.
- **≥1000 kcal/d extra → fat accrual.**
- Inadequate calories → protein is burned instead of spared for fetal growth. Extra needs may be offset by ↓ physical activity.

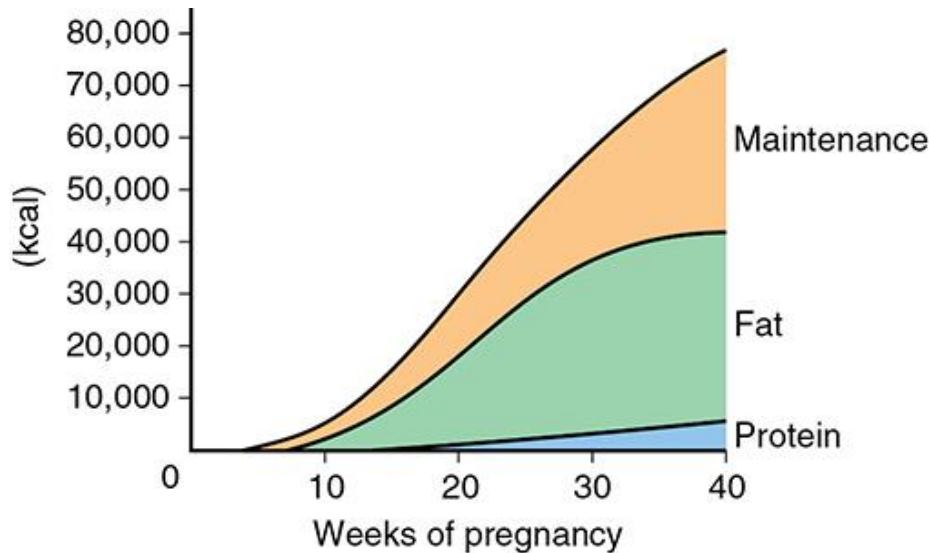


Figure 10-4. Cumulative kcal cost of pregnancy (~77,000–80,000 kcal by 40 wk): maintenance and fat make up most of it; protein is a small share; most accrues after 20 wk.

Protein

- **~1000 g** deposited in the second half (**5–6 g/d**) → intake **~1 g/kg/d** (may need doubling late in gestation).
- Most plasma amino acids **fall** (ornithine, glycine, taurine, proline); **glutamic acid and alanine rise**.
- Best from **animal sources**; milk and dairy are nearly ideal (protein + calcium).

Minerals

- Adequate-calorie diets supply enough minerals **except iron and iodine**.
- **Iron:**
 - ~300 mg to fetus + placenta, 500 mg to maternal Hb mass, nearly all **after midpregnancy** → **~7 mg/d**.
 - Supplement **≥27 mg elemental iron daily** (in most prenatal vitamins). **30 mg/d** (ferrous gluconate, sulfate or fumarate) in the second half meets pregnancy **and lactation** needs.
 - Labels: mg of compound followed by **mg of elemental iron in parentheses**.
 - **60–100 mg/d** if: **obese, multifetal, late start, irregular intake, or low Hb**.
 - Overt iron deficiency responds well to oral iron; **ferritin rises more than Hb**.
- **Iodine:** **220 µg/d**; iodized salt and bread. Intake has ↓ over 15 years. Severe deficiency → **endemic cretinism**; very early supplementation prevents some cases.
- **Calcium:** **~30 g** retained, mostly to the fetus late; only **~2.5%** of maternal calcium. **Routine calcium does not prevent preeclampsia**.
- **Zinc:** **~12 mg/d**; severe deficiency → poor appetite, poor growth, impaired healing. Supplement only zinc-deficient women in poor-resource countries. Vegetarians have lower intake.
- **Magnesium:** deficiency not a feature of pregnancy (seen after **intestinal bypass**). Supplementation (365 mg, 13–24 wk) did **not** improve outcomes.
- **Selenium:** deficiency → often fatal **cardiomyopathy**; toxicity from oversupplementation; not needed in US women.
- **Potassium:** plasma ↓ **~0.5 mEq/L** by midpregnancy; deficiency e.g. with **hyperemesis**.
- **Fluoride:** no added benefit from maternal fluoride if the newborn gets fluoridated water; not transferred into breast milk.

Vitamins

- A diet adequate in calories and protein supplies most vitamins, **except folic acid** in times of unusual need (**protracted vomiting, hemolytic anemia, multiple fetuses**).
- Multivitamins in poor countries ↓ LBW and FGR but not preterm delivery or perinatal mortality.

Folic acid

- Fortification (1998) cut NTD-affected pregnancies from **~4000** → **~3000/year**.
- **400 µg/d periconceptionally** could prevent **more than half** of NTDs. Fortification (140 µg/100 g grain) adds only ~100 µg/d, so **supplements are still needed**.
- **Prior child with NTD**: recurrence **2–5%** → **4 mg/d** from the **month before conception through the 1st trimester** cuts it by **>70%**. Take as a **separate folic acid supplement, not extra multivitamins** (avoids excess fat-soluble vitamins).

Other vitamins

- **Vitamin A**: **>10,000 IU/d** is teratogenic (like **isotretinoin**). **β-carotene** does not cause toxicity. US intake adequate → no routine supplement. Deficiency (developing world) → **night blindness**, maternal anemia, spontaneous preterm birth.
- **Vitamin B12**: plasma ↓ in pregnancy (↓ transcobalamins). Only in animal foods → **strict vegetarians'** infants have low stores, worse with breastfeeding. Excess **vitamin C** → functional B12 deficiency. B12 deficiency may be an independent NTD risk factor.
- **Vitamin B6 (pyridoxine)**: not routinely needed; **2 mg/d** if nutrition is poor. With **doxylamine** for nausea and vomiting.
- **Vitamin C**: **80–85 mg/d** (~20% above nonpregnant); diet suffices. Maternal levels ↓, cord levels higher (true of most water-soluble vitamins).
- **Vitamin D**: deficiency common (limited sun, vegetarians, **darker skin**) → congenital rickets, neonatal fractures. **Adequate intake 15 µg/d (600 IU/d)**. Measure **25-hydroxyvitamin D** if deficiency suspected (optimal level unknown). Supplementation in asthmatic mothers may ↓ childhood asthma.

Pragmatic nutritional surveillance

1. Eat what she wants in reasonable amounts, **salted to taste**
2. Ensure food is available to deprived women
3. Monitor weight gain against **IOM** goals
4. Periodic **dietary recall** to catch errant diets
5. **≥30 mg elemental iron daily**; **folate** before and early in pregnancy; **iodine** where deficient
6. Recheck **Hct/Hb at 28–32 weeks**

6. Common Concerns

Employment

- **Family and Medical Leave Act (1993)**: up to **12 weeks unpaid leave**.
- Uncomplicated pregnancy → can usually work **until labor**.
- **Preterm birth risk slightly ↑** with standing/walking **>3 h/day**, lifting **>5 kg**, or physical exertion. Working nulliparas had a **5-fold ↑ preeclampsia** (Higgins, 2002). Avoid severe physical strain and undue fatigue.

Exercise

- Generally **no need to limit** exercise, if not excessively fatigued and no injury risk. Some studies show larger placentas and birthweights; working women who exercised had smaller neonates and more dysfunctional labor (Magann).
- **ACOG: ≥150 minutes/week of moderate-intensity activity** (after clinical evaluation, if no contraindication). Uterine artery Doppler is not affected.
- **Safe**: walking, running, swimming, stationary cycling, low-impact aerobics.
- **Avoid**: high risk of **falls or abdominal trauma**; **scuba diving** (fetal decompression sickness).
- Rest may benefit: hypertensive disorders, preterm labor, **placenta previa**, severe cardiac/pulmonary disease, **multiples, FGR**.

Table 10-6. Some Contraindications to Exercise During Pregnancy

Category	Examples
Significant cardiovascular or pulmonary disease	Chest pain, calf pain or swelling
Significant risk for preterm labor	Cerclage, multifetal gestation, significant bleeding, threatened preterm labor, ruptured membranes
Obstetrical complications	Preeclampsia, placenta previa, anemia, poorly controlled diabetes or epilepsy, morbid obesity, fetal-growth restriction

Summarized from American College of Obstetricians and Gynecologists, 2017g; The American Academy of Pediatrics, 2017.

Seafood consumption

- Fish = protein, low saturated fat, **omega-3 fatty acids**.
- **Eat 8–12 oz of fish weekly**, but **≤6 oz albacore ("white") tuna**.
- **Avoid high-mercury fish: shark, swordfish, king mackerel, tile fish.**
- Local fish with unknown mercury → **≤6 oz/week total**.
- No raw/undercooked fish (**listeriosis**).

Lead screening

- Lead → gestational hypertension, miscarriage, LBW, neurodevelopmental impairment.
- Test **only if a risk factor** is present. **>5 µg/dL** → find and remove source, repeat levels. **>45 µg/dL** = lead poisoning → consider **chelation**, with expert consultation.

Automobile and air travel

- **Three-point restraint:** lap belt **under the abdomen across the upper thighs**, shoulder belt **between the breasts. Do not disable airbags.**
- **Air travel is safe up to 36 weeks** if uncomplicated. Keep seatbelt on; **support stockings, leg movement, hourly ambulation** to ↓ VTE. Main risks: infection and distance from care.

Coitus

- Usually **not harmful**. **Avoid** with threatened **miscarriage, placenta previa or preterm labor**.
- By 36 wk, **72%** have intercourse <1×/week. Late-pregnancy intercourse does **not** trigger labor (no ↑, even ↓ delivery within 2 weeks).
- ⚠️ **Air blown into the vagina during cunnilingus** → **fatal air embolism** has been reported.

Dental care

- Periodontal disease is **linked to preterm labor**, but treating it does **not** prevent preterm birth.
- Caries are not aggravated by pregnancy. **Dental treatment and dental radiographs are not contraindicated.**

Immunization

- **Thimerosal**–neuropsychological disorder link is **groundless**.
- **Influenza and Tdap are recommended for ALL pregnant women.**
- Rubella-susceptible women → **MMR postpartum** (contraindicated in pregnancy).

Table 10-7. Recommendations for Immunization During Pregnancy and Postpartum

Immunobiological Agent	Indications for Immunization During Pregnancy	Dose Schedule	Comments
Live Attenuated Virus Vaccines			
Measles	Contraindicated—see immune globulins	Single dose SC, preferably as MMR ^a	Vaccinate susceptible women postpartum; breastfeeding is not a contraindication
Mumps	Contraindicated	Single dose SC, preferably as MMR	Vaccinate susceptible women postpartum
Rubella	Contraindicated, but congenital rubella syndrome has never been described after vaccine	Single dose SC, preferably as MMR	Teratogenicity of vaccine is theoretical and not confirmed to date; vaccinate susceptible women postpartum
Immunobiological Agent	Indications for Immunization During Pregnancy	Dose Schedule	Comments
Poliomyelitis (oral = live attenuated; injection = enhanced-potency inactivated virus)	Not routinely recommended for women in the United States, except women at increased risk of exposure ^b	Primary: Two doses of enhanced-potency inactivated virus SC at 4- to 8-week intervals and a 3rd dose 6–12 months after 2nd dose. Immediate protection: One dose oral polio vaccine (in outbreak setting)	Vaccine indicated for susceptible women traveling in endemic areas or in other high-risk situations
Yellow fever	Travel to high-risk areas	Single dose SC	Limited theoretical risk outweighed by risk of yellow fever
Varicella	Contraindicated, but no adverse outcomes reported in pregnancy	Two doses needed: 2nd dose given 4–8 weeks after 1st dose	Teratogenicity of vaccine is theoretical; vaccination of susceptible women should be considered postpartum
Zoster	Contraindicated	Single dose	Teratogenicity is theoretical
Immunobiological Agent	Indications for Immunization During Pregnancy	Dose Schedule	Comments
Smallpox (vaccinia)	Contraindicated in pregnant women and in their household contacts	One dose SC, multiple pricks with lancet	Only vaccine known to cause fetal harm
Other			
Influenza	All pregnant women, regardless of trimester during flu season (October–May)	One dose IM every year	Inactivated or recombinant virus vaccine
Rabies	Indications for prophylaxis not altered by pregnancy; each case considered individually	Public health authorities to be consulted for indications, dosage, and route of administration	Killed-virus vaccine

Immunobiological Agent	Indications for Immunization During Pregnancy	Dose Schedule	Comments
Human papillomavirus	Not recommended	Three-dose series IM at 0, 1, and 6 months	Nonavalent vaccine is inactivated virus; no teratogenic effects have been observed
Hepatitis B	Preexposure and postexposure for women at risk of infection, e.g., chronic liver or kidney disease	Three-dose series IM at 0, 1, and 6 months	Used with hepatitis B immune globulin for some exposures; exposed newborn needs birth-dose vaccination and immune globulin as soon as possible; all infants should receive birth dose of vaccine
Hepatitis A	Preexposure and postexposure if at risk (international travel); chronic liver disease	Two-dose schedule IM, 6 months apart	Inactivated virus

Inactivated Bacterial Vaccines

Immunobiological Agent	Indications for Immunization During Pregnancy	Dose Schedule	Comments
Pneumococcus	(1) For chronic metabolic, liver, cardiac, or lung ds. (2) For immunosuppression, general malignancy, chronic renal disease, or asplenia	One lifetime dose: (1) One dose PPSV23; (2) One dose PCV13 with PPSV23 given 8 weeks later	Indications not altered by pregnancy. Polyvalent polysaccharide vaccine; safety in the first trimester has not been evaluated
Meningococcus	Indications not altered by pregnancy; vaccination recommended in unusual outbreaks	One dose MenACWY or MPSV4; tetravalent vaccine; two doses for asplenia	Antimicrobial prophylaxis if significant exposure
Typhoid	Not recommended routinely except for close, continued exposure or travel to endemic areas	Killed. Primary: 2 injections IM 4 weeks apart. Booster: One dose; schedule not yet determined	Killed, injectable vaccine or live attenuated oral vaccine; oral vaccine preferred
Anthrax	See text	Six-dose primary vaccination, then annual booster vaccination	Preparation from cell-free filtrate of <i>B anthracis</i> ; no dead or live bacteria; teratogenicity of vaccine theoretical

Immunobiological Agent	Indications for Immunization During Pregnancy	Dose Schedule	Comments
Toxoids			
Tetanus-diphtheria-acellular pertussis (Tdap)	Recommended in every pregnancy, preferably between 27 and 36 weeks to maximize passive antibody transfer	Primary: Two doses IM at 1–2 month interval with 3rd dose 6–12 months after the 2nd. Booster: Single dose IM every 10 years, as a part of wound care if ≥ 5 years since last dose, or once per pregnancy	Combined tetanus-diphtheria toxoids with acellular pertussis (Tdap) preferred; updating immune status should be part of antepartum care
Specific Immune Globulins			
Hepatitis B	Postexposure prophylaxis	Depends on exposure	Usually given with hepatitis B virus vaccine; exposed newborn needs immediate prophylaxis

Immunobiological Agent	Indications for Immunization During Pregnancy	Dose Schedule	Comments
Rabies	Postexposure prophylaxis	Half dose at injury site, half dose in deltoid	Used in conjunction with rabies killed-virus vaccine
Tetanus	Postexposure prophylaxis	One dose IM	Used in conjunction with tetanus toxoid
Varicella	Should be considered for exposed pregnant women to protect against maternal, not congenital, infection	One dose IM within 96 hours of exposure	Indicated also for newborns or women who developed varicella within 4 days before delivery or 2 days following delivery
Standard Immune Globulins			

Immunobiological Agent	Indications for Immunization During Pregnancy	Dose Schedule	Comments
Hepatitis A (hepatitis A virus vaccine should be used with hepatitis A immune globulin)	Postexposure prophylaxis and those at high risk	0.02 mL/kg IM in one dose	Immune globulin should be given as soon as possible and within 2 weeks of exposure; infants born to women who are incubating the virus or are acutely ill at delivery should receive one dose of 0.5 mL as soon as possible after birth

^aTwo doses necessary for students entering institutions of higher education, newly hired medical personnel, and travel abroad. ^bInactivated polio vaccine recommended for nonimmunized adults at increased risk. Ds. = disease; ID = intradermally; IM = intramuscularly; MMR = measles, mumps, rubella; PO = orally; SC = subcutaneously. From ACOG 2018o; CDC 2011; Kim, 2016; Munoz, 2019.

Caffeine

- **~500 mg/d (about 5 cups)** slightly ↑ miscarriage; **<200 mg/d** ("moderate") shows no ↑.
- Preterm birth and fetal growth: unclear. CARE study: **>200 mg/d → 1.4-fold FGR** vs <100 mg/d.
- **ACOG: <200 mg/d is not associated with miscarriage or preterm birth**; the FGR link is unsettled.

Nausea and heartburn

- **Nausea/vomiting**: starts between the **1st and 2nd missed period**, lasts to **14–16 wk**.
 - Affects **¾** of women, average **35 days**. **Half relieved by 14 wk, 90% by 22 wk**. All-day in **80%** ("morning sickness" is a misnomer).
 - Treatment: **small frequent meals, ginger**; mild → **vitamin B6 + doxylamine**; some need **phenothiazine or H1-blocker** antiemetics.
 - **Hyperemesis gravidarum** → dehydration, electrolyte/acid–base disturbance, starvation ketosis (Ch 57).
- **Heartburn**: reflux from upward displacement of the stomach + **relaxed LES**. Avoid bending or lying flat; small frequent meals; **antacids** (aluminum hydroxide, magnesium trisilicate, magnesium hydroxide).

Pica and ptyalism

- **Pica** prevalence **~30%** worldwide (4% in one US 2nd-trimester survey).
 - **Pagophagia** (ice), **amyllophagia** (starch), **geophagia** (clay).
 - May be triggered by **iron deficiency**, but not all have it; pica itself worsens iron deficiency.
 - Most common items: **starch 64%**, dirt 14%, sourdough 9%, ice 5%.
- **Ptyalism**: profuse salivation; sometimes after starch ingestion; common with **hyperemesis gravidarum**.

Headache or backache

- New-onset/new-type headache in **≥5%** of pregnancies → **acetaminophen** for most.

- **Low back pain in ~70%**; worse with advancing gestation, obesity and a prior history.
 - Prevent: **squat rather than bend**, back-support pillow, **avoid high heels**.
 - **Severe pain needs orthopedic evaluation** (pregnancy-associated osteoporosis, disc disease, vertebral osteoarthritis, septic arthritis).
 - Treatment: analgesics, heat, rest; acetaminophen; **NSAIDs only in short courses**; muscle relaxants (cyclobenzaprine, baclofen); then physical therapy; **sacroiliac support belt**.

Varicosities and hemorrhoids

- **Femoral venous pressure (supine): 8 → 24 mm Hg** at term → leg varicosities worsen, especially standing.
 - Treat with rest + **leg elevation, elastic stockings**; surgery generally not advised in pregnancy.
 - Superficial varicosities are a **risk factor for DVT/PE**.
- **Vulvar varicosities**: may be massive; **rupture → severe bleeding**. Specially fitted pantyhose; foam-rubber pad on a belt.
- **Hemorrhoids** in **up to 40%**: topical anesthetics, warm soaks, stool softeners. **Thrombosed external hemorrhoid** → incise and remove the clot under local anesthesia.

Sleeping and fatigue

- Fatigue early (**progesterone** soporific effect + nausea); later discomfort, frequency, dyspnea.
- **Sleep-disordered breathing** → hypertensive disorders, stillbirth, preterm delivery.
- 3rd trimester: ↓ sleep efficiency, more awakenings, ↓ **stage 4 and REM** sleep.
- Daytime naps; mild sedative such as **diphenhydramine** at bedtime.

Cord blood banking

- Hemopoietic stem cells treat **>70 diseases**.

	Public bank	Private bank
Use	Allogeneic donation (related or unrelated recipient)	Autologous future use
Cost	Free donation	Fees for processing and annual storage

- If asked, explain the pros and cons of each. Transplants from cord blood stored **without a known indication are rare**; use for the donor family is remote.
- **Directed donation** is recommended when an immediate family member has a condition treatable by hemopoietic transplantation.

Quick-Recall: Prenatal Timeline

Timing	Test / action
8–9 days post-ovulation	hCG detectable (doubles every 1.4–2 days)
4–5 wk	Gestational sac on TVS
Mid-5th wk	Yolk sac (confirms IUP)
>6 wk	Embryo with cardiac motion
10 wk	FHR by Doppler; cfDNA possible from here; hCG peak (60–70 days)
First visit	History, exam, Pap, CBC, type/Rh/antibody screen, HIV, HBsAg, syphilis, rubella, urine culture, chlamydia (all); gonorrhea/TB if at risk
10–14 wk	1st-trimester aneuploidy screen
15–20 wk	2nd-trimester serum screen / NTD screen
16–18 wk (multip) / ~2 wk later (primip)	Quickening
20 wk	Fetoscope FHR in 80% (all by 22 wk); fundal height ≈ weeks (20–34 wk)
24–28 wk	GDM testing
27–36 wk	Tdap (every pregnancy)
28 wk	Repeat antibody screen in D-negative → anti-D immunoglobulin
28–32 wk	Repeat Hb/Hct (± syphilis)
3rd trimester (<36 wk)	Retest HIV, syphilis, chlamydia, gonorrhea in high-risk women
35–37 wk	GBS vaginal–rectal culture
≤36 wk	Air travel allowed
Any time, flu season	Inactivated influenza vaccine
Postpartum	MMR, varicella for susceptible women

Key Numbers to Memorize

- No prenatal care: **1.6%** of US births; **5-fold** ↑ maternal death
- hCG doubling **1.4–2.0 days**; peak **60–70 days**; plateau **16 wk**; home test needs **12.5 mIU/mL** for 95% at missed period
- Visits: q**4 wk** to 28, q**2 wk** to 36, then **weekly**; WHO model **5 vs 8** visits
- Naeglele: LMP **+7 d –3 mo**; pregnancy **280 d**; trimesters **14/28/42 wk**
- Uterus: small orange **6 wk**, large orange **8 wk**, grapefruit **12 wk**
- FHR **110–160**; Doppler **10 wk**; stethoscope **20 wk (80%)**, **22 wk (all)**; fundal height = weeks at **20–34 wk**; FGR missed in **1/3**
- Weight gain (lb): underweight **28–40**, normal **25–35**, overweight **15–25**, obese **11–20**; twins normal BMI **37–54**
- Postpartum: **5.4 kg** lost at delivery; **~1 kg** retained
- Calories: **80,000 kcal** total; **+0/340/452 kcal/d** by trimester; protein **71 g/d (~1 g/kg)**
- Iron **27 mg/d** (≥30 mg elemental; **60–100 mg** if high need); iodine **220 µg**; folate **600 µg** (RDA), **400 µg** supplement, **4 mg** after prior NTD; vitamin D **600 IU**; vitamin A teratogenic **>10,000 IU**
- Calcium retained **30 g** (2.5% of maternal); K⁺ ↓ **0.5 mEq/L**
- Chlamydia **2–13%**; test of cure **3–4 wk**; IPV **4–8%**; smoking **7.2%**; alcohol **12%**; illicit drugs **10%**; methadone start **10–30 mg/d**
- Exercise **≥150 min/wk**; fish **8–12 oz/wk**, albacore **≤6 oz**; lead **>5 / >45 µg/dL**; caffeine **<200 mg/d**
- Nausea: **¾** of women, 50% gone by **14 wk**, 90% by **22 wk**; back pain **70%**; hemorrhoids **40%**; pica **30%**
- Femoral venous pressure **8 → 24 mm Hg**; VZIG within **96 h**; FMLA **12 wk**